[Affiliation]

October 29, 2026

Editor-in-Chief
Advanced Engineering Informatics

Dear Editor,

Please find enclosed our manuscript "Coverage-controlled learning of connection-invariant magnetic surrogates for reconfigurable electric machines" for consideration as a research paper in Advanced Engineering Informatics.

Reconfigurable windings are an established route to wide-speed-range, high-efficiency electrified powertrains, yet candidate connections are usually evaluated with dedicated FE models or flux-map tables. Our manuscript addresses this with one idea: a winding connection is a set of linear constraints on coil currents, so a single connection-invariant surrogate—trained once on coil-level FE data—can query unseen connections; coverage determines when zero-shot transfer is reliable. We think this fits the journal's scope in three specific ways: (i) it is an engineering-informatics method (learned surrogates embedded in design and operation workflows) rather than a new machine or a new controller; (ii) it comes with a reproducible benchmark—seven hold-out connections, three difficulty tiers, 9,276+ nonlinear FE samples and fixed splits—released with code and data; and (iii) it delivers actionable engineering findings, including two data-collection principles, a quantified physical-consistency advantage of co-energy surrogates, and an honest negative drive-cycle result. A 105-point voltage/gear sweep found no case in which reconfiguration created feasibility. The direct-MLP screen predicts gains only in constrained designs (28 cases, 1–2.64 %); the nominal schedule is energy-neutral within 0.006 %. An independent FEMM cross-check limits the claims explicitly: it supports torque trends on three unseen connections ($R^2 = 0.86$–0.94) while showing that pointwise torque remains mesh-sensitive; an apparently low-energy tap75 design failed its FEMM torque check.

All results in the manuscript are reproducible from the released code and data with one command per figure; the FE reference data, trained models, and benchmark splits are included. The work has not been published elsewhere and is not under consideration elsewhere. The authors declare no competing interests.

We suggest the following potential reviewers with relevant expertise: [3–6 suggested reviewers with affiliations and emails].

Thank you for your consideration.

Sincerely,

[Author Name]